



Pythagoras' theorem in context

- 1 Izzy writes down a correct Pythagoras' theorem equation for a triangle. Which of the three sides of this triangle is the hypotenuse? Tick **1** correct answer

$$a^2 = 13^2 + 19^2$$

side A side B side C

- ☒ side A
☐ side B
☐ side C
☐ impossible to tell

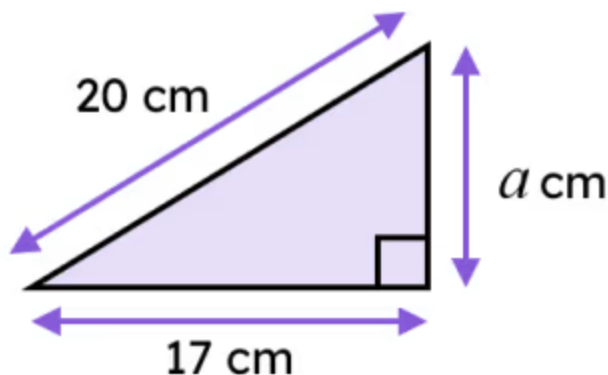
- 2 Aisha rearranges a correct Pythagoras' theorem equation for a triangle. Which of these statements are correct? Tick **2** correct answers

$$181^2 - 19^2 = w^2$$

side Y side X side W

- ☐ The hypotenuse of the triangle is side W
☐ The hypotenuse of the triangle is side X
☒ The hypotenuse of the triangle is side Y
☐ Side W has length of 32 400 units.
☒ Side W has a length of 180 units.

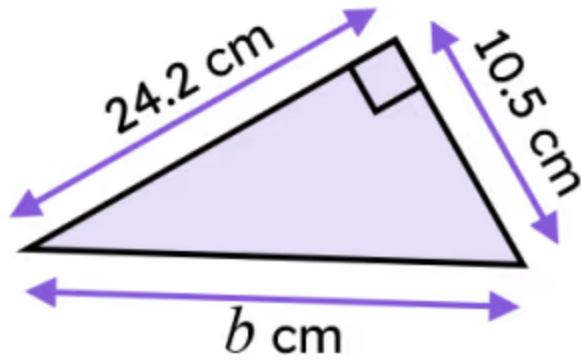
- 3 Which of these are correct for this triangle? Tick **3** correct answers



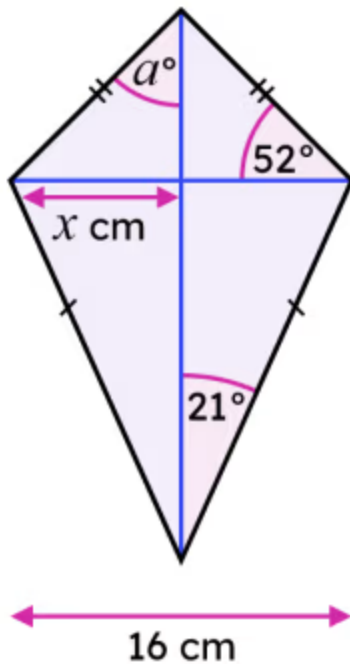
- ☐ $20^2 + a^2 = 17^2$
☒ $20^2 = a^2 + 17^2$
☒ $a^2 = 20^2 - 17^2$
☒ $a = 10.5$ (to 1 d.p.)
☐ $a = 26.4$ (to 1 d.p.)



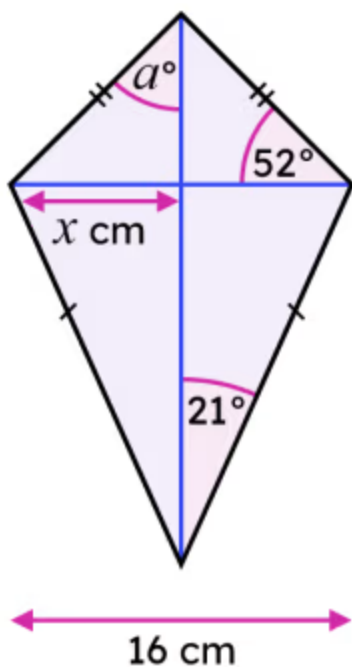
- 4 What is the value of b , rounded to 2 d.p.? $b = \underline{26.38}$. Fill in the blank



- 5 The diagonals of this kite have been drawn on. The value of x is 8. Fill in the blank



- 6 The diagonals of this kite have been drawn on. Which of these statements are correct. Tick 2 correct answers



- ☐ The two diagonals of a kite intersect to create two acute angles of equal length
- ☒ The two diagonals of a kite intersect at right angles
- ☒ $a^\circ = 38^\circ$
- ☐ $a^\circ = 69^\circ$
- ☐ $a^\circ = 73^\circ$

